# Practice quiz: Regression

**Total points** 2

**1.** **1 point**

For linear regression, the model is $f_{w,b}(x) = wx + b$.

Which of the following are the inputs, or features, that are fed into the model and with which the model is expected to make a prediction?

- (●) $x$
- ( ) $m$
- ( ) $(x, y)$
- ( ) $w$ and $b$.

**2.** For linear regression, if you find parameters $w$ and $b$ so that $J(w, b)$ is very close to zero, what can you conclude? **1 point**

- (●) The selected values of the parameters $w$ and $b$ cause the algorithm to fit the training set really well.
- ( ) The selected values of the parameters $w$ and $b$ cause the algorithm to fit the training set really poorly.
- ( ) This is never possible -- there must be a bug in the code.

# Practice quiz: Supervised vs unsupervised learning

**Total points** 2

**1.** Which are the two common types of supervised learning? (Choose two) **1 point**

- [x] Classification
- [ ] Clustering
- [x] Regression

**2.** **1 point**

Which of these is a type of unsupervised learning?

- ( ) Regression
- ( ) Classification
- (●) Clustering

# Practice quiz: Train the model with gradient descent

**Total points** 2

**1.**

1 point

Gradient descent is an algorithm for finding values of parameters w and b that minimize the cost function J.

repeat until convergence {

$$w = w - \alpha \frac{\partial}{\partial w} J(w, b)$$

$$b = b - \alpha \frac{\partial}{\partial b} J(w, b)$$

When $\frac{\partial J(w,b)}{\partial w}$ is a negative number (less than zero), what happens to $w$ after one update step?

- ○ $w$ stays the same
- ○ It is not possible to tell if $w$ will increase or decrease.
- ○ $w decreases$
- ◉ $w$ increases.

**2.**

1 point

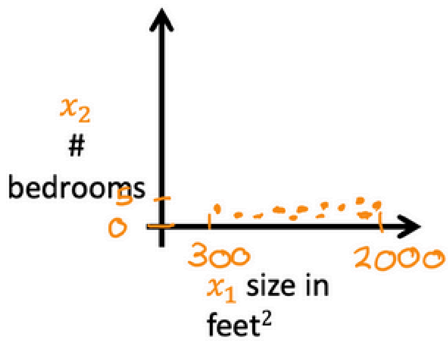
For linear regression, what is the update step for parameter b?

- ○ $b = b - \alpha \frac{1}{m} \sum\limits_{i=1}^{m} (f_{w,b}(x^{(i)}) - y^{(i)}) x^{(i)}$
- ◉ $b = b - \alpha \frac{1}{m} \sum\limits_{i=1}^{m} (f_{w,b}(x^{(i)}) - y^{(i)})$

# Practice quiz: Gradient descent in practice

**Total points** 4

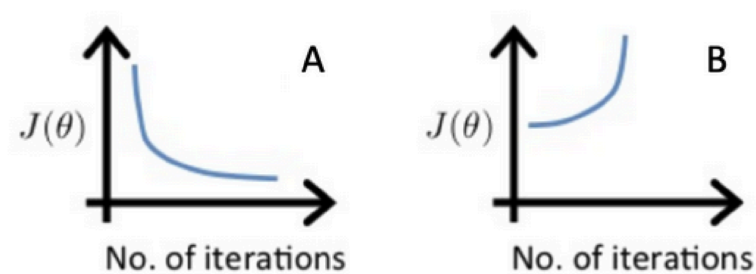
**1.** 1 point




Which of the following is a valid step used during feature scaling?

- (●) Subtract the mean (average) from each value and then divide by the (max - min).
- ( ) Add the mean (average) from each value and and then divide by the (max - min).

**2.** Suppose a friend ran gradient descent three separate times with three choices of the learning rate $\alpha$ and plotted the learning curves for each (cost J for each iteration). 1 point




For which case, A or B, was the learning rate $\alpha$ likely too large?

- ( ) Both Cases A and B
- (●) case B only
- ( ) Neither Case A nor B
- ( ) case A only

**3.** Of the circumstances below, for which one is feature scaling particularly helpful?

1 point

- (●) Feature scaling is helpful when one feature is much larger (or smaller) than another feature.
- ( ) Feature scaling is helpful when all the features in the original data (before scaling is applied) range from 0 to 1.

**4.**

1 point

You are helping a grocery store predict its revenue, and have data on its items sold per week, and price per item. What could be a useful engineered feature?

- (●) For each product, calculate the number of items sold times price per item.
- ( ) For each product, calculate the number of items sold divided by the price per item.

**3.** True/False? To make gradient descent converge about twice as fast, a technique that almost always works is to double the learning rate $alpha$.

1 point

- ( ) True
- (●) False

**4.**

1 point

True/False? With polynomial regression, the predicted values f_w,b(x) does not necessarily have to be a straight line (or linear) function of the input feature x.

- (●) True
- ( ) False

1. Consider the problem of predicting how well a student does in her second year of college/university, given how well they did in their first year.

   Specifically, let x be equal to the number of "A" grades (including A-. A and A+ grades) that a student receives in their first year of college (freshmen year). We would like to predict the value of y, which we define as the number of "A" grades they get in their second year (sophomore year).

   Refer to the following training set of a small sample of different students' performances (note that this training set will also be referenced in other questions in this quiz). Here each row is one training example. Recall that in linear regression, our hypothesis is $h_\theta(x) = \theta_0 + \theta_1 x$, and we use $m$ to denote the number of training examples.

| x | y |
|---|---|
| 3 | 4 |
| 2 | 1 |
| 4 | 3 |
| 0 | 1 |

   For the training set given above, what is the value of $m$? In the box below, please enter your answer (which should be a number between 0 and 10).

   4

2. For this question, continue to assume that we are

   using the training set given above. Recall our definition of the

   cost function was $J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})^2$.

   What is $J(0,1)$? In the box below,

   please enter your answer (use decimals instead of fractions if necessary, e.g., 1.5).

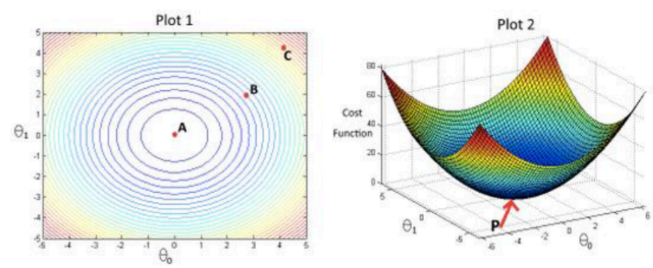
   0.5

3. Suppose we set $\theta_0 = 0, \theta_1 = 1.5$. What is $h_\theta(2)$?

   3

4. In the given figure, the cost function $J(\theta_0, \theta_1)$ has been plotted against $\theta_0$ and $\theta_1$, as shown in 'Plot 2'. The contour plot for the same cost function is given in 'Plot 1'. Based on the figure, choose the correct options (check all that apply).

Plots for Cost Function J($\theta_0$,$\theta_1$)




- [ ] If we start from point B, gradient descent with a well-chosen learning rate will eventually help us reach at or near point A, as the value of cost function $J(\theta_0, \theta_1)$ is maximum at point A.
- [x] If we start from point B, gradient descent with a well-chosen learning rate will eventually help us reach at or near point A, as the value of cost function $J(\theta_0, \theta_1)$ is minimum at A.
- [x] Point P (the global minimum of plot 2) corresponds to point A of Plot 1.
- [ ] If we start from point B, gradient descent with a well-chosen learning rate will eventually help us reach at or near point C, as the value of cost function $J(\theta_0, \theta_1)$ is minimum at point C.
- [ ] Point P (The global minimum of plot 2) corresponds to point C of Plot 1.

5. Suppose that for some linear regression problem (say, predicting housing prices as in the lecture), we

have some training set, and for our training set we managed to find some $\theta_0$, $\theta_1$ such that $J(\theta_0, \theta_1) = 0$. Which

of the statements below must then be true? (Check all that apply.)

- [ ] For this to be true, we must have $y^{(i)} = 0$ for every value of $i = 1, 2, \ldots, m$.
- [ ] For this to be true, we must have $\theta_0 = 0$ and $\theta_1 = 0$

  so that $h_\theta(x) = 0$
- [x] Our training set can be fit perfectly by a straight line,

  i.e., all of our training examples lie perfectly on some straight line.
- [ ] Gradient descent is likely to get stuck at a local minimum and fail to find the global minimum.

2.

You run gradient descent for 15 iterations

with $\alpha = 0.3$ and compute $J(\theta)$ after each

iteration. You find that the value of $J(\theta)$ **increases** over

time. Based on this, which of the following conclusions seems

most plausible?

- ○ $\alpha = 0.3$ is an effective choice of learning rate.
- ○ Rather than use the current value of $\alpha$, it'd be more promising to try a larger value of $\alpha$ (say $\alpha = 1.0$).
- ◉ Rather than use the current value of $\alpha$, it'd be more promising to try a smaller value of $\alpha$ (say $\alpha = 0.1$).